

Syed A. Ali

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EDUCATION

Vanderbilt University

August 2022 - May 2026

M.S. & B.S. in Computer Science; GPA: 3.99/4.00

Nashville, TN

B.A. in Mathematics, Minor in Physics

- **Relevant Coursework:** Distributed Systems, Software Engineering, Machine Learning, Artificial Intelligence, Deep Learning, Algorithms, Data Structures, Operating Systems, Intermediate Software Design, Cloud Computing

SKILLS

Languages: Typescript, Python, C++, Java, JavaScript, Rust, C#, SQL, MongoDB

Technologies: Node.js, AWS (Lambda, DynamoDB, S3), React, Next.js, GraphQL, PyTorch, Playwright, JUnit

Skills: AGILE, CI/CD, Project Planning, Kubernetes, Docker, Unit Testing, Git, WSL, CMake, Figma, Github Copilot

WORK EXPERIENCE

Capital One

June 2025 - August 2025

Software Engineer, Internship

Richmond, VA

- Enhanced an enterprise-level cloud automation platform by developing key full-stack features to improve user adoption and data visibility for over **1,400** internal teams.
- Engineered a real-time GitHub activity display with Node.js and GraphQL and built user-facing components with React, utilizing an **AGILE** workflow to impact the management of over **11,000** automations.
- Constructed automated failover solutions using AWS (ECS, Step Functions, DynamoDB), improving platform resilience for **130+** internal teams while ensuring all new code met a minimum of **80%** unit test coverage.
- Secured **3rd place** in a company-wide intern hackathon by developing a phishing training tool that simulated attacks and tracked failure rates by capturing clicked links, built with Next.js, Prisma, SQLite, and Amazon SES.

American Bureau of Shipping

May 2024 - August 2024

Software Engineer, Internship

Houston, TX

- Built a Graph Neural Network with PyTorch Geometric to predict next-port destinations, improving model accuracy to **85%+**, and cutting costs by **15%** and using data from **1200+** ports.
- Achieved **97%** accuracy in classifying machine part failures for Maersk using TF-IDF, word embeddings, and models like Random Forest, XGBoost, and MLP.

Vanderbilt Department of Transportation

September 2023 - January 2025

Transportation Software Engineer

Nashville, TN

- Spearheaded the analysis of **1.7 million** transit data entries using **Python (Pandas, Numpy)**, optimizing data retrieval by **28%** with **Oracle Analytics Cloud** and the **Nashville MTA**.
- Created an internal Plotly dashboard using Python with SQL integration, reducing data analysis time by **34%**.
- Launched the public-facing MoveVU Dashboard using Next.js, TypeScript, and PostgreSQL, displaying ridership insights of over **10,000** monthly users.

Vanderbilt Department of Computer Science

August 2023 - Present

Graduate Teaching Assistant

Nashville, TN

- Assisted **10+** students weekly in software design ('25), operating systems ('24), and data structures ('23).
- Converted **3000+** lines of legacy testing code to Java using the JUnit framework.

PROJECTS

Access911 | Next.js, AWS (Lambda, DynamoDB, Bedrock), TypeScript, Twilio, Mapbox

October 2025

- Secured **3rd place** and **\$10,000** at the **AWS x Vanderbilt Mission Autonomy Hackathon** by developing an AI-powered emergency response platform that reduces 911 hold times during disasters.
- Architected serverless AWS infrastructure using Lambda and DynamoDB for concurrent call processing; integrated Amazon Bedrock (Claude Sonnet 3.5) for real-time call summarization.
- Built end-to-end voice pipeline (Twilio, ElevenLabs) and Next.js dispatcher dashboard with live Mapbox visualization, achieving **93% success rate** and 2.8s latency across **2,500 calls** in 30 minutes.

CourseCoach | Next.js, TypeScript, OpenAI API, Chakra UI

November 2024

- Developed and presented a course scheduling platform, placing **2nd at VandyHacks XI**, enabling students to optimize academic planning through AI-powered recommendations and interactive flowcharts.
- Integrated OpenAI API to provide personalized course recommendations into Next.js web app based on user history and preferences, improving decision-making and ensuring degree progress.